Forecasting Foreign Exchange Reserves For Sri Lanka : A multivariate time series approach

Abstract

This paper compares classical econometric, Bayesian, regime-switching, and machine-learning models for forecasting Sri Lanka's foreign exchange reserves over 2005–2025, including the 2022 sovereign default episode. Using rolling-origin evaluation across five variable sets, we find that Markov-Switching VAR in a disaggregated balance-of-payments specification delivers the strongest point-forecast performance, reducing RMSE by roughly three-quarters relative to a random-walk benchmark in the post-default window. A factorial decomposition shows that model architecture explains more of this gain than simply expanding the information set, and that balance-of-payments disaggregation is most effective when paired with regime-switching structure. These results support regime-aware, component-level reserve forecasting as a practical framework for policy surveillance in crisis-prone emerging markets.

Keywords: Foreign exchange reserves, forecasting, Markov-switching VAR, balance of payments disaggregation, regime dynamics, machine learning, Sri Lanka, sovereign default, emerging markets

JEL Classification: C52, C53, E47, F31, F37

1 Introduction

For emerging market economies, adequate foreign exchange reserves serve as the primary defence against balance-of-payments crises, currency runs, and sovereign debt distress [Jeanne and Rancière, 2011]. Sri Lanka's experience between 2019 and 2022 underscores this point. Gross reserves declined from approximately US$7.6 billion at end-2019 to an estimated US$50 million of usable reserves by April 2022, precipitating the country's first sovereign default since independence [Athukorala, 2024]. GDP contracted by more than 7%, inflation surpassed 50%, and widespread social upheaval toppled the sitting government [Weerakoon and Jayasuriya, 2023].

Despite this policy relevance, reserve forecasting has attracted considerably less academic attention than exchange rate prediction, inflation forecasting, or GDP nowcasting

[Rossi, 2013]. Existing contributions are largely confined to single-model frameworks [Sehgal and Sharma, 2008, Gosselin and Parent, 2005] and rarely assess performance across structural breaks. No existing study benchmarks classical, Bayesian, regime-switching, and machine learning approaches on identical data with formal statistical comparison, or evaluates forecast accuracy through a sovereign default episode.

This paper has three objectives. First, it compares the out-of-sample accuracy of classical econometric, Bayesian, regime-switching, and machine learning models for reserve forecasting, evaluated on a dataset spanning stable accumulation, pandemic disruption, sovereign default, and IMF-supervised recovery. Second, it decomposes the relative contributions of model architecture and information content through a factorial design. Third, it provides a component-level scenario framework for policy surveillance, with quantified sensitivity to currency, trade, and programme disbursement shocks.

The analysis evaluates twelve model specifications across five variable sets of increasing dimensionality, using rolling-origin backtests with expanding estimation windows. A temporally motivated train-validation-test split isolates pre-crisis (through 2019), crisis (2020–2022), and post-default (2023–2025) regimes. Model comparisons are formalised through Diebold and Mariano [1995] tests and the Model Confidence Set of Hansen et al. [2011].

The main findings are as follows. MS-VAR achieves RMSE reductions of 74–77% relative to the naïve benchmark in the parsimonious and BoP specifications. A factorial decomposition shows that model architecture accounts for the bulk of this gain. Aggregating balance-of-payments flows into net flows destroys approximately 94% of the gross flow signal. The Meese-Rogoff puzzle does not hold in its strong form: MS-VAR outperforms the random walk even during the acute crisis segment. Diagnostic analysis of regime persistence, impulse response asymmetry, and classification certainty provides analytical evidence for the sources of this advantage.

The paper contributes to three areas. Empirically, it provides the first comparison of classical, Bayesian, regime-switching, and machine learning reserve forecasting approaches evaluated through a sovereign default, addressing a gap in the literature where forecast accuracy has been tested only during stable periods. Methodologically, it introduces a factorial decomposition of architecture and information effects and a formal test for information loss under flow aggregation. For policy, it contributes a scenario analysis framework applicable to central bank reserve management and IMF surveillance.

2 Background and Literature Review

2.1 From Adequacy Ratios to Dynamic Forecasting

The reserve adequacy literature has produced a series of static threshold metrics: import-cover rules, the Guidotti-Greenspan ratio [Greenspan, 1999], monetary-based benchmarks [Calvo, 1996, Wijnholds and Kapteyn, 2001], and the IMF's composite Assessing Reserve Adequacy (ARA) framework [IMF, 2011, 2015]. Obstfeld et al. [2010] shifted attention to financial-sector liabilities as the relevant scale variable, while Jeanne and Rancière [2011] formalised the cost-benefit calculus of reserve holdings.

These frameworks share a fundamental limitation: they assess reserve *levels* against static benchmarks rather than forecasting reserve *trajectories*. Static ratios cannot anticipate the speed of depletion, the timing of threshold breaches, or the state-dependent dynamics that accelerate crises. Early warning systems [Kaminsky et al., 1998, Frankel and Saravelos, 2012] partially address this gap by predicting binary crisis events, but they do not produce the continuous reserve path forecasts needed for operational policy planning. Dynamic forecasting models are therefore a necessary complement to adequacy ratios, particularly in settings where reserves can decline rapidly across regime boundaries.

Two further features of reserve dynamics complicate static approaches. First, reserves are actively managed policy variables, not passive residuals of balance-of-payments accounting. Aizenman and Lee [2007] distinguished precautionary motives (buffering against sudden stops) from mercantilist motives (supporting export competitiveness), each generating different accumulation dynamics and depletion vulnerabilities. Second, the marginal value of reserves is highly nonlinear: negligible when buffers are large, but very high near the critical zone where market confidence collapses [Jeanne and Rancière, 2011]. This threshold nonlinearity implies that linear models are misspecified near crisis points, because the relationship between reserves and their determinants changes qualitatively with the reserve level itself.

2.2 Reserve Forecasting Literature

Table 1 summarises the key contributions to the reserve forecasting literature. Several patterns are evident. First, the literature is sparse relative to exchange rate or inflation forecasting. Second, nearly all studies evaluate performance during stable accumulation periods; none systematically assesses accuracy through a sovereign default. Third, the literature predominantly treats reserves as a scalar aggregate, ignoring the compositional dynamics of underlying balance-of-payments flows.

Sehgal and Sharma [2008] applied ARIMA models to Indian reserves and found acceptable in-sample fit but limited out-of-sample reliability. A key limitation is the univariate

Table 1: Summary of Reserve Forecasting Literature

Study	Country	Method(s)	Crisis?	Key Finding
Sehgal and Sharma [2008]	India	ARIMA	No	Reasonable in-sample fit; limited out-of-sample reliability
Gosselin and Parent [2005]	12 EM	Panel ECM	No	Long-run relationships hold; model breaks down in crises
Gupta et al. [2014]	China	DMA, DMS, BMA	No	DMS outperforms linear models; time-varying weights matter more than predictor expansion
Salas et al. [2025]	102 countries	BMA-Naïve Bayes	No	Strong classification accuracy; no sovereign default evaluation
This paper	Sri Lanka	MS-VAR, BVAR, XGBoost, DMA, + others	Yes	Regime-switching dominates; architecture matters more than information

Notes: EM = Emerging Markets. "Crisis?" indicates whether the evaluation window includes a sovereign default or comparable episode.

framework, which cannot capture the multivariate dynamics of balance-of-payments components. Gosselin and Parent [2005] extended the approach to a panel error-correction framework across twelve economies, documenting cointegrating relationships between reserves, imports, and short-term debt. However, their model broke down during crisis episodes, highlighting the central challenge that forecast accuracy is most needed when it is hardest to achieve. Gupta et al. [2014] applied DMA and DMS to Chinese reserves and found that time-varying model weights outperform both linear models and static model averaging. Their result that adaptive model composition matters more than predictor expansion motivates the inclusion of DMA and DMS in the present study. Salas et al. [2025] applied BMA across 102 countries with strong classification results, but did not evaluate performance through a sovereign default.

Machine learning methods have shown competitive results in adjacent forecasting problems [Medeiros et al., 2021, Richardson et al., 2021], yet their systematic application to reserve forecasting remains limited. The broader evidence suggests that machine learning offers its greatest advantage in stable periods with strong nonlinear patterns, while simpler approaches often dominate during structural breaks [Richardson et al., 2021].

2.3 Methodological Foundations

The Meese-Rogoff result [Meese and Rogoff, 1983] established that structural exchange rate models cannot outperform a random walk out of sample. This finding has proved robust across currencies and time periods [Rossi, 2013, Cheung et al., 2005]. Reserves are driven by many of the same balance-of-payments fundamentals and exhibit similar nonlinearities. Taylor et al. [2001] argued that nonlinear mean reversion becomes detectable when deviations from fundamentals are large, suggesting that the puzzle may weaken during strong directional trends.

The Markov-switching framework [Hamilton, 1989, Krolzig, 1997] is particularly suited to this setting because it allows all model parameters to shift between discrete states. Peria [2002] and Brunetti et al. [2007] applied Markov-switching models to speculative attacks in European and Southeast Asian settings, but neither applied the framework to reserve forecasting with disaggregated balance-of-payments flows. Bayesian VARs with Minnesota priors [Litterman, 1986, Banbura et al., 2010] address the curse of dimensionality through informative shrinkage, while DMA and DMS [Raftery et al., 2010, Koop and Korobilis, 2012] allow model weights to evolve over time. Gradient-boosted trees [Chen and Guestrin, 2016] and LSTM networks [Hochreiter and Schmidhuber, 1997] offer flexible nonlinear alternatives, though their performance is sensitive to hyperparameter choices and training set size.

Three gaps in this literature motivate the present study. First, no existing comparison jointly evaluates classical, Bayesian, regime-switching, and machine learning approaches

for reserve forecasting on identical data and evaluation criteria. Second, regime-switching models have not been applied to disaggregated balance-of-payments components in a reserve forecasting context. Third, systematic evaluation through a sovereign default is absent. Gupta et al. [2014] demonstrated the value of DMA and DMS for reserve forecasting with linear constituent models; the present study extends their approach by incorporating regime-switching specifications into the DMA/DMS model pool, testing whether dynamic combination of regime-aware and linear forecasts yields further gains.

Based on the theoretical considerations and empirical gaps identified above, this paper tests three hypotheses. (H1) Regime-switching models outperform linear and machine learning alternatives for reserve forecasting in a sample that includes a sovereign default. (H2) Model architecture (regime-switching versus linear/tree-based) explains more of the forecast accuracy differential than information content (parsimonious versus disaggregated variable sets). (H3) Aggregating gross balance-of-payments flows into net flows destroys predictive information through cancellation of offsetting components.

2.4 The Sri Lankan Reserve Crisis

Sri Lanka's gross reserves declined from approximately US$7.6 billion at end-2019 to an estimated US$50 million of usable reserves by April 2022, when the government suspended external debt service [Athukorala, 2024]. The decline was driven by tourism revenue loss (COVID-19 and the 2019 Easter attacks), fiscal deterioration, and a Central Bank policy of defending the pegged exchange rate through reserve sales. Between September 2021 and March 2022 alone, gross reserves fell from US$3.1 billion to under US$2 billion. An IMF Extended Fund Facility (US$2.9 billion) was approved in March 2023, and by early 2024 gross reserves had rebuilt to approximately US$4.4 billion [Weerakoon and Jayasuriya, 2023, UNDP, 2022].

This chronology generates a reserve series with at least three distinct regimes: gradual accumulation (2005–2019), rapid depletion (2020–2022), and post-default rebuilding (2023 onward). Each regime exhibits different dynamics, volatility, and relationships to driving variables. Bai-Perron multiple structural break tests applied to the gross reserves series identify five significant breaks (2009:10, 2014:04, 2017:05, 2020:02, 2022:04), confirming the regime structure that motivates the emphasis on Markov-switching specifications.

3 Data and Variable Construction

3.1 Data Sources and Sample

The dataset comprises nine monthly time series spanning January 2005 to December 2025 (252 observations), drawn from the Central Bank of Sri Lanka (CBSL) statistical database, the IMF's International Reserves and Foreign Currency Liquidity (IRFCL) template, the Colombo Stock Exchange market statistics, and Sri Lanka Customs trade returns. All monetary variables are denominated in US dollars to ensure comparability and to avoid conflating real dynamics with exchange rate valuation effects. Table 2 reports descriptive statistics for all series.

Table 2: Descriptive Statistics of Key Variables

Variable	Obs	Mean	Std Dev	Min	Max
Gross Reserves (USD m)	252	5450.6	2105.4	1588.4	9935.8
Exports (USD m)	224	886.8	176.8	282.3	1302.2
Imports (USD m)	225	1487.3	315.8	606.3	2241.0
Trade Balance (USD m)	224	−601.1	227.8	−1100.6	−39.1
Remittances (USD m)	202	507.8	121.8	204.9	812.7
Tourism (USD m)	192	171.4	128.0	0.0	475.2
Exchange Rate (LKR/USD)	228	177.0	77.3	107.6	363.3
M2 (USD m)	225	25436.4	12829.8	7095.5	48091.1
CSE Net Flows (USD m)	166	3.6	26.3	−62.2	143.8

Sample period: 2005-01 to 2025-12 (252 months). Exchange rate expressed as LKR per USD.

3.2 Dependent Variable

Gross official reserves (USD millions) constitute the primary dependent variable, serving as the numerator of every major adequacy benchmark [Jeanne and Rancière, 2011, Obstfeld et al., 2010]. Joint ADF and KPSS testing confirms that gross reserves are integrated of order one. The first-differenced series (reserve change) is therefore the primary forecast target for the regime-switching systems. For MS-VAR and MS-VECM, predicted changes are mapped back to levels by cumulative summation from the last observed reserve level. The naïve benchmark is a random walk in levels, $\hat{y}_{t+h|t} = y_t$ for all h. All reported point metrics (RMSE, MAE, MAPE, MASE) are computed on level forecasts against observed levels for every model, including the benchmark, ensuring an identical target scale. The first-differenced series exhibits a negative first-order autocorrelation (ACF(1) $= -0.301$), indicating mean-reverting intervention behaviour that differs between accumulation and crisis regimes.

Official reserve data from the CBSL statistical database and the IMF's IRFCL template are subject to minimal measurement error relative to forecast uncertainty, as they reflect audited central bank balance sheet positions. We do not model measurement error explicitly but note that any such error would attenuate estimated forecast accuracy differences, making the reported performance gaps conservative.

3.3 Variable Set Design

Five variable sets of increasing dimensionality are defined to disentangle the effect of model specification from the effect of information content. This design is motivated by the well-documented tension in small-sample macroeconomic forecasting between information gains from additional predictors and estimation error from parameter proliferation [Banbura et al., 2010, Carriero et al., 2015]. By evaluating each model across all five sets, the analysis identifies whether forecast failures stem from model misspecification or from over- or under-parameterisation.

Table 3: Variable Set Specifications and Effective Samples

Variable Set	Key Variables	k	Effective Coverage
Parsimonious	Reserves, trade balance, exchange rate	3	2007:01–2025:12
BoP	Reserves, exports, imports, remittances, tourism	5	2009:01–2025:03
Monetary	Reserves, exchange rate, M2	3	2005:01–2025:06
PCA	Reserves + 3 PCs from 8 indicators	4	2012:01–2024:11
Full	All available predictors	9	2012:01–2025:03

Notes: k = number of endogenous variables in the VAR system. Missing-data handling follows a uniform rule before evaluation: forward-fill up to 3 consecutive months (matching the quarterly reporting frequency of several source series), then drop remaining missing rows. No model-specific imputation is used in the horse-race stage. The BoP set replaces trade balance with separate exports and imports to identify independent shocks. The PCA set extracts three principal components from eight indicators, fitted on training data only to avoid look-ahead bias.

The **Parsimonious set** ($k = 3$: reserves, trade balance, exchange rate) captures the first-order determinants of reserve dynamics under the import-cover and exchange rate defence frameworks. The trade balance, computed as FOB exports minus CIF imports, serves as the aggregate current account proxy. The USD/LKR exchange rate is fundamentally endogenous to reserve dynamics under Sri Lanka's managed float, since central bank intervention to defend the peg directly depletes reserves [Obstfeld et al., 2010].

The **Balance of Payments set** ($k = 5$) disaggregates the current account into exports, imports, remittances, and tourism earnings alongside reserves. The disaggregation is driven by the requirements of the MS-VAR framework, which needs to identify independent shocks to each flow component. This is critical because BoP components exhibit

sharply divergent crisis dynamics: during the 2020–2022 period, exports and imports moved in opposite directions, whilst remittances and tourism followed distinct structural break patterns.

The **Monetary set** ($k = 3$) comprises reserves, the exchange rate, and broad money (M2), introducing the domestic capital flight channel identified by Obstfeld et al. [2010] and captured in the IMF's ARA metric. This compact specification tests whether monetary aggregates provide incremental forecasting power beyond trade flows.

The **PCA set** ($k = 4$) comprises reserves plus three principal components extracted from the eight indicators in the broader dataset. This tests whether dimensionality reduction can preserve informational content while reducing estimation burden. The approach is particularly relevant for the BVAR and MS-VAR specifications, where parameter proliferation in small samples is a binding constraint. The three retained components explain 79.5% of total variance (PC1: 46.1%, PC2: 20.8%, PC3: 12.7%).

The **Full set** ($k = 9$) includes all available predictors simultaneously. It serves as an upper bound on available information and as a test of whether regularisation mechanisms (Minnesota shrinkage, tree-based feature selection, LSTM dropout) can effectively manage the dimensionality.

Table 3 highlights the ragged-edge implications of uneven source coverage: variable sets are estimated on different effective windows, but all models within a given variable set are evaluated on the same preprocessed sample. Cross-model comparisons within a variable set are therefore sample-aligned; cross-variable-set comparisons should be interpreted with the effective-sample caveat in mind.

3.4 Stationarity and Seasonal Properties

All series are subjected to joint ADF and KPSS testing to determine the order of integration. Variables confirmed as I(1) are first-differenced for VAR estimation; where cointegrating relationships are identified via the Johansen procedure, the VECM specification is estimated in levels with error-correction terms. Seasonal properties are assessed via STL decomposition. Variables exhibiting seasonal strength above 0.5—notably tourism earnings (0.593)—are deseasonalised using X-13ARIMA-SEATS prior to estimation in specifications that do not include seasonal dummies.

4 Methodology

4.1 Model Overview

The model set spans classical econometrics (ARIMA, VAR, VECM), regime-switching specifications (MS-VAR, MS-VECM), Bayesian approaches (BVAR), dynamic model

combination (DMA, DMS), machine learning (XGBoost, LSTM, XGB-Quantile), and naïve benchmarks (random walk, seasonal naïve). Additional specifications include a balance-of-payments identity model (BoPIdentity) and a local-level stochastic volatility model (LocalLevelSV). All models are estimated in an expanding-window fashion, initialising on the training period (January 2005–December 2019, 180 observations) and expanding monthly through the validation (2020–2022) and test (2023–2025) periods. Standard estimation details are reported in Appendix A.

This section focuses on the non-standard modelling choices that distinguish the present application from textbook implementations.

4.2 Regime-Switching Specifications

The MS-VAR allows all parameters to shift between two discrete regimes (accumulation and depletion) governed by an unobserved Markov chain with state $S_t \in \{0, 1\}$:

$$\mathbf{y}_t = \mathbf{c}_{S_t} + \sum_{i=1}^{p} \mathbf{A}_{i,S_t} \mathbf{y}_{t-i} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{S_t})$$

Parameters are estimated via the EM algorithm. The key modelling choice is the two-regime specification, motivated by the distinct accumulation and depletion dynamics visible in the reserve series. Higher-order specifications ($K = 3$) were explored but yielded poorly identified regimes with short durations, consistent with the limited within-regime observations available in a 252-month sample. At the default iteration cap, some full-sample runs can stop at the maximum iteration bound; accordingly, the regime diagnostics are complemented with a seeded multi-start stability check under a higher iteration cap (Section 6.2).

For the MS-VECM extension, a critical design decision concerns the cointegrating vectors:

$$\Delta \mathbf{y}_t = \boldsymbol{\alpha}_{S_t} \boldsymbol{\beta}' \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_{i,S_t} \Delta \mathbf{y}_{t-i} + \mathbf{u}_t$$

The cointegrating vectors $\boldsymbol{\beta}$ are estimated on the full sample and held regime-invariant, whilst the adjustment speeds $\boldsymbol{\alpha}_{S_t}$ and short-run dynamics $\boldsymbol{\Gamma}_{i,S_t}$ are regime-dependent. This follows the approach of Krolzig [1997]: the long-run equilibrium between reserves, trade flows, and the exchange rate is assumed to hold across regimes, but the speed at which the system corrects deviations from equilibrium differs between accumulation and crisis states.

4.3 XGBoost Rolling Feature Engineering

The XGBoost implementation departs from standard applications through its rolling feature engineering pipeline, which constructs 3-month and 6-month rolling means and volatility estimates for all variables at each forecast origin. This creates a feature set that evolves with the expanding estimation window, capturing momentum and volatility clustering that are economically meaningful for reserve dynamics. The rolling construction ensures that no future information leaks into the feature set. XGB-Quantile extends the framework to produce probabilistic forecasts by predicting the 10th, 25th, 50th, 75th, and 90th percentiles of the conditional distribution. Time-series cross-validation tuning is employed for XGBoost in the parsimonious and BoP specifications, though robustness is mixed across folds.

4.4 Architecture-vs-Information Factorial Decomposition

A key methodological contribution is the formal disentangling of architecture (model class) from information content (variable set). We construct a 2×2 factorial design crossing model class $\mathcal{M} \in \{\text{MS-VAR}, \text{XGBoost}\}$ with information set $\mathcal{I} \in \{\text{Parsimonious}, \text{BoP}\}$. For RMSE outcomes $R_{m,i}$, the architecture effect (holding information constant) is:

$$\hat{\alpha} = \frac{1}{2} \sum_{i \in \mathcal{I}} \left[R_{\text{XGB},i} - R_{\text{MSVAR},i} \right].$$

The information effect (holding model constant) is:

$$\hat{\iota} = \frac{1}{2} \sum_{m \in \mathcal{M}} \left[R_{m,\text{Parsim}} - R_{m,\text{BoP}} \right].$$

The interaction term captures whether the effect of information depends on architecture:

$$\hat{\delta} = \left(R_{\text{XGB,Parsim}} - R_{\text{XGB,BoP}} \right) - \left(R_{\text{MSVAR,Parsim}} - R_{\text{MSVAR,BoP}} \right).$$

This decomposition follows a standard two-factor analysis of variance structure applied to out-of-sample RMSE. It quantifies whether the MS-VAR advantage is driven primarily by regime-switching architecture, by richer information in disaggregated flows, or by the interaction between the two. The decomposition is descriptive rather than causal: it isolates the marginal contribution of each factor within the observed comparison set, but does not identify treatment effects in the econometric sense.

4.5 Dynamic Model Averaging and Selection

Dynamic Model Averaging and Dynamic Model Selection [Raftery et al., 2010, Koop and Korobilis, 2012] address model uncertainty by allowing the weights on competing forecasts to evolve over time. At each forecast origin t, DMA computes a posterior model probability $\pi_{t|t-1,k}$ for each constituent model k and forms the combined forecast as the probability-weighted average. DMS selects the single model with the highest $\pi_{t|t-1,k}$. The forgetting factor α governs how quickly past performance is discounted. We set $\alpha = 0.99$, following the convention in Koop and Korobilis [2012], implying an effective memory of roughly 100 periods. The rolling forecast-error variance is estimated over a 24-month window, chosen to span approximately two years of monthly data and thus capture medium-term shifts in relative model performance.

The constituent model pool comprises all seven non-DMA/DMS models evaluated on the BoP specification: ARIMA, BVAR, MS-VAR, MS-VECM, Naïve, VECM, and XGBoost. This design allows DMA and DMS to shift weight toward regime-switching models during crisis periods and toward simpler specifications during tranquil periods, an adaptive strategy that static model selection cannot replicate.

4.6 Information Loss Under Aggregation

To test whether aggregating balance-of-payments components into composite variables destroys predictive information, we formalise the aggregation mapping. Let $\mathbf{X}_t = (x_{1t}, \ldots, x_{kt})'$ denote the vector of gross BoP flows and $A_t = \mathbf{w}'\mathbf{X}_t$ the scalar aggregate (e.g., trade balance = exports − imports). Forecasting with A_t conditions on a strictly coarser σ-field than forecasting with $\mathbf{X}_t$, implying potential information loss. We quantify the empirical severity of this loss through a cancellation index:

$$\mathrm{CI}_t = \frac{|A_t|}{\sum_{j=1}^{k} |x_{jt}|},$$

which measures the ratio of the net flow to the sum of absolute gross flows. When CI_t is near zero, large gross flows cancel almost entirely, and the aggregate retains almost no information about the underlying dynamics.

4.7 Evaluation Framework

Forecasts at horizons $h \in \{1, 3, 6, 12\}$ months are evaluated using RMSE, MAE, MAPE, and MASE [Hyndman and Koehler, 2006] for point accuracy. Density forecasts are evaluated using CRPS and prediction interval coverage probability (PICP) at the 80% and 95% levels. An asymmetric policy loss function penalises under-prediction of reserve depletion at twice the rate of over-prediction (2:1 weight ratio), motivated by the observation

that under-prediction of depletion carries higher policy costs than over-prediction, since responses to false alarms are inexpensive relative to the cost of undetected crises. All metrics are computed on reserve levels, with the random-walk benchmark transformed identically to the structural models.

Statistical significance is assessed via the Diebold and Mariano [1995] test of equal predictive accuracy (with Harvey-Leybourne-Newbold small-sample correction) and the Model Confidence Set of Hansen et al. [2011]. The MCS uses a stationary block bootstrap with 1,000 replications and block length $\lceil T^{1/3} \rceil$, where T is the number of forecast observations. Formal definitions of all metrics are provided in Appendix A.

5 Empirical Results

5.1 Main Forecasting Comparison

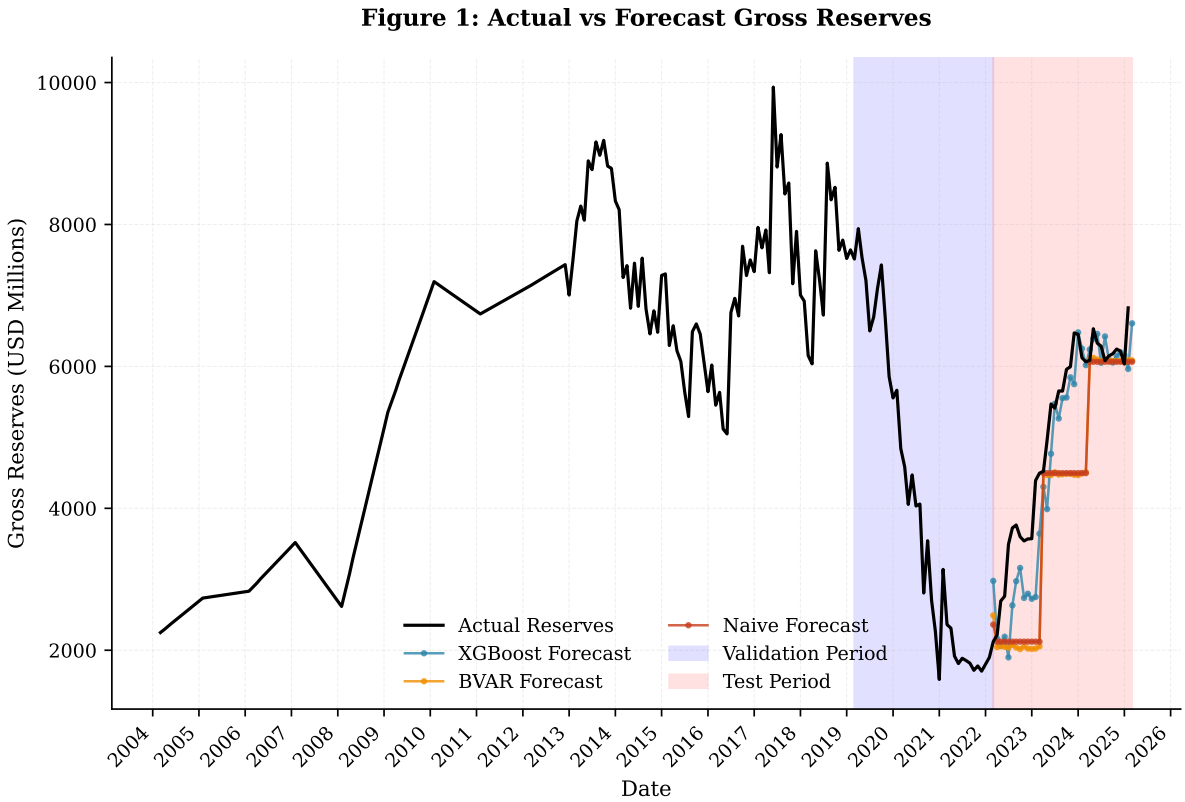

Figure 1: Actual gross reserves and one-step-ahead forecasts from selected models. The validation period (2020–2022, blue shading) covers the pandemic and sovereign default; the test period (2023–2025, red shading) covers the IMF-supervised recovery.

Table 4 presents out-of-sample forecast accuracy for the parsimonious specification at $h = 1$ over the test period (2023:01–2025:12). MS-VAR achieves RMSE of 311.8, versus 1,178.9 for the naïve random walk, a 73.6% improvement. This advantage is consistent across all metrics: MAE of 230.3 (versus 935.9), MAPE of 5.20% (versus 20.47%), and

Table 4: Out-of-Sample Forecast Accuracy: Parsimonious Specification, $h = 1$

Model	RMSE	MAE	MAPE	MASE	Policy Loss	CRPS	n
MS-VAR	**311.8**	**230.3**	**5.20**	**0.77**	**408.4**	–	36
MS-VECM	357.7	280.5	6.31	0.94	512.7	–	36
XGBoost	640.6	475.7	11.82	1.59	900.8	466.4	36
XGB-Quantile	655.5	521.8	12.25	1.75	1018.0	241.0	36
ARIMA	1170.4	896.2	21.61	3.00	1785.6	734.2	36
Naïve	1178.9	935.9	20.47	3.14	1864.2	761.5	36
VECM	1194.5	956.8	21.64	3.21	1892.3	782.4	36
BVAR	1214.9	964.4	21.41	3.23	1917.2	792.1	36

Notes: MAPE in %. Policy Loss = asymmetric loss (2× penalty for under-prediction). CRPS = Continuous Ranked Probability Score. Bold indicates best performance. Test period: 2023:01–2025:12. Dashes indicate models without probabilistic output in this specification.

MASE of 0.77, which falls below the 1.0 threshold indicating improvement over the naïve in-sample benchmark. MS-VECM follows at 357.7.

XGBoost achieves RMSE of 640.6. This is better than all classical and Bayesian specifications, but roughly double the MS-VAR error. Classical models (ARIMA, VECM) cluster in the range 1,170–1,195, performing no better than the naïve benchmark. BVAR achieves 1,214.9, reflecting the difficulty of estimating a high-dimensional system with limited pre-crisis training data.

Two distinct performance tiers emerge: regime-switching models (RMSE 312–358) and all others (RMSE > 640). This gap motivates the diagnostic investigation in Section 6.

5.2 Statistical Significance

Table 5: Diebold-Mariano Test Results: Selected Model Pairs

Model Pair	DM p-value	Significance
MS-VAR vs. ARIMA	0.0006	***
MS-VAR vs. BVAR	<0.0001	***
MS-VAR vs. VECM	<0.0001	***
MS-VAR vs. Naïve	<0.0001	***
MS-VAR vs. XGBoost	0.0012	**
MS-VECM vs. XGBoost	0.0033	**
XGBoost vs. ARIMA	0.0052	**

Notes: Diebold-Mariano test with Harvey-Leybourne-Newbold small-sample correction. Two-sided p-values. ***: $p < 0.001$; **: $p < 0.01$. Pairwise matrix totals cited in the text refer to the nine-model BoP specification, where the effective $h = 1$ test window is 2023:01–2025:03 ($n = 27$).

Table 5 reports Diebold-Mariano tests for key model pairs. MS-VAR's advantage over all non-regime-switching, non-combination models is highly significant ($p < 0.01$ in every case). The MS-VAR vs. XGBoost comparison yields $p = 0.0012$, confirming that the regime-switching advantage is not a statistical artefact. Across the full nine-model set in the BoP specification (including DMA and DMS), 30 of 36 pairwise DM comparisons are significant at the 10% level, and 28 at the 1% level.

The Model Confidence Set at $\alpha = 0.10$ retains four models: MS-VAR, DMA, MS-VECM, and DMS. All five remaining models are eliminated: XGBoost ($p = 0.009$), ARIMA, Naïve, BVAR, and VECM (all $p < 0.001$). The MCS does not single out one winner. Instead, it draws a boundary between regime-aware methods (including dynamic combinations that lean heavily on regime-switching constituents) and all other specifications.

These inference results should be interpreted with appropriate caution. The BoP test window contains 27 effective monthly observations at $h = 1$. Overlapping multi-step horizons further reduce effective independence (roughly $n_{\text{eff}} \approx n/h$: about 9 at $h = 3$, 4–5 at $h = 6$, and 2–3 at $h = 12$). In short samples, both DM asymptotics and block-bootstrap MCS outcomes are sensitive to tuning choices such as block length. We therefore treat p-values as complementary evidence and place primary weight on rank stability and economic magnitude. In the BoP $h = 1$ test window, MS-VAR lowers RMSE from 1,350.8 (naïve) to 315.2, a reduction of 1,035.6 USD million, approximately 22.3% of mean test-period reserves.

5.3 Horizon Sensitivity

Table 6: Forecast Accuracy Across Horizons: RMSE (Parsimonious Specification)

Model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	h_{12}/h_1
XGBoost	**640.6**	**1281.7**	1887.6	2664.2	4.16
Naïve	1178.9	1393.2	**1737.4**	**2373.7**	2.01
VECM	1194.5	1404.6	1751.5	2398.4	2.01
BVAR	1214.9	1438.0	1793.5	2488.9	2.05
XGB-Quantile	655.5	1384.5	1991.1	3095.0	4.72
ARIMA	1170.4	1561.6	2453.0	3444.6	2.94

Notes: h_{12}/h_1 = deterioration ratio from 1-month to 12-month horizon. Bold indicates best performance at each horizon. Test period: 2023–2025. Parsimonious MS-VAR and MS-VECM horizon results are omitted here to avoid duplication with Table 8, which reports the preferred BoP regime-switching specification used for the main regime comparisons.

Table 6 reveals pronounced horizon dependence among non-regime-switching models. XGBoost dominates at short horizons ($h = 1$: 640.6; $h = 3$: 1,281.7). But this advantage deteriorates. By $h = 6$, the naïve random walk (1,737.4) overtakes XGBoost (1,887.6).

The pattern reflects a limitation of machine learning in macroeconomic forecasting: the model exploits patterns in the recent training distribution, and distributional shift erodes that advantage at longer horizons. Bayesian methods maintain more stable relative performance. BVAR's deterioration ratio is 2.05 versus 4.16 for XGBoost, reflecting the regularising benefit of the Minnesota prior.

5.4 Validation versus Test Period

Table 7: Forecast Performance: Crisis Validation vs. Post-Default Test Period

	Validation (2020–2022)			Test (2023–2025)		
Model	RMSE	MAE	MAPE	RMSE	MAE	MAPE
XGBoost	1183.5	1037.3	37.82	640.6	475.7	11.82
XGB-Quantile	1055.3	884.8	32.81	655.5	521.8	12.25
ARIMA	1986.3	1723.7	70.36	1170.4	896.2	21.61
Naïve	1328.1	1029.3	35.69	1178.9	935.9	20.47
VECM	1532.2	1289.6	46.57	1194.5	956.8	21.64
BVAR	1412.7	1128.9	39.59	1214.9	964.4	21.41

Notes: Validation period coincides with the Sri Lankan economic crisis and sovereign default (April 2022). Test period covers the post-default recovery under the IMF Extended Fund Facility. MAPE in %. This table focuses on the non-regime-switching parsimonious set; MS-VAR and MS-VECM period splits are reported in the BoP-focused regime tables to avoid repetitive presentation across specifications.

Table 7 stratifies results by the crisis validation period (2020:01–2022:12) and post-default test period (2023:01–2025:12). During the acute crisis, MAPE ranges from 33% to over 70% across these non-regime-switching models, indicating severe forecasting difficulty. In sharp contrast, during the post-crisis recovery, XGBoost achieves MAPE of 11.82%, and multiple models achieve MAPE in the 20–22% range. However, the regime-switching models—which are not shown in this parsimonious-only comparison but dominate the BoP results below—substantially outperform even during the crisis segment, as discussed in Section 6.6.

5.5 Balance of Payments Specification Results

Table 8 reveals a tight cluster of regime-aware models at the top of the BoP specification. MS-VAR achieves RMSE of 315.2 at $h = 1$—a 76.7% improvement over the naïve benchmark (1,350.8). DMA follows at 321.7, MS-VECM at 323.4, and DMS at 326.0. The gap separating these four models from the next competitor is large: XGBoost, at 886.1, trails by more than 560 RMSE points. At longer horizons, DMS becomes increasingly competitive ($h = 6$: 1,049.8), and LSTM emerges as the strongest performer at $h = 12$

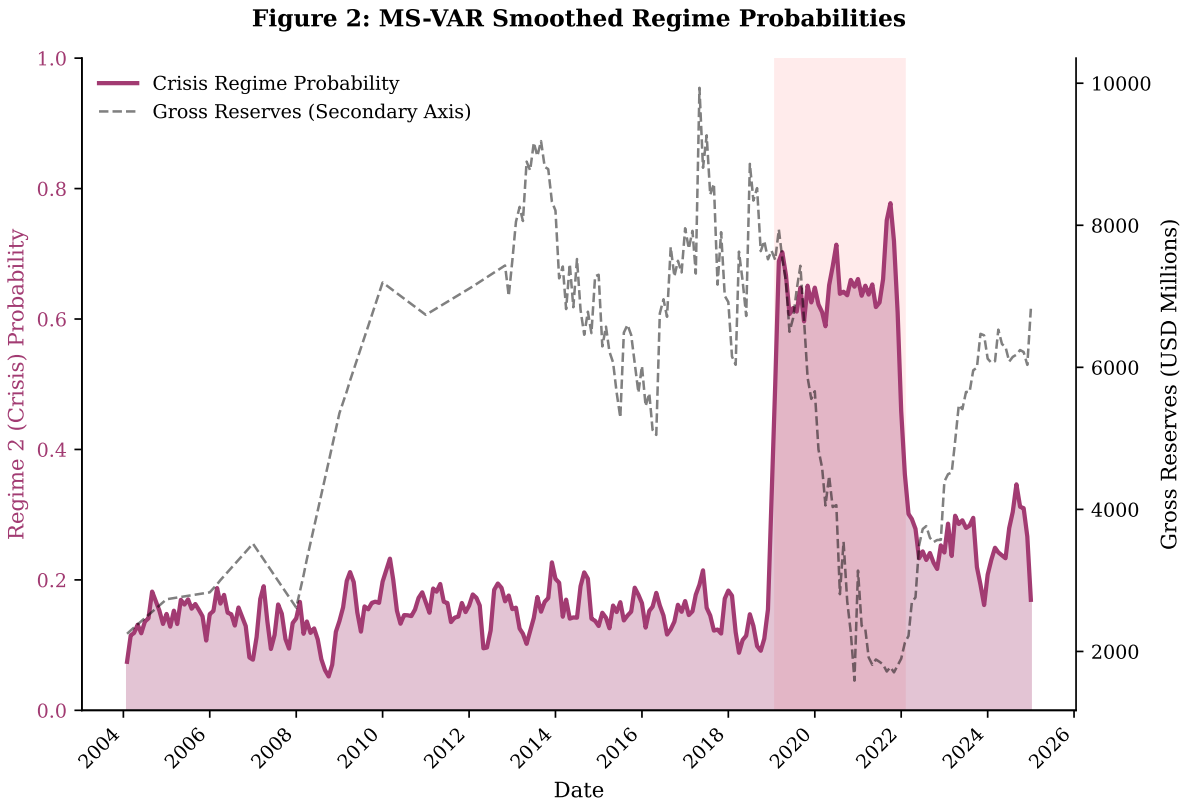

Figure 2: MS-VAR smoothed regime probabilities (top axis) and actual gross reserves (bottom axis). The estimated crisis regime probability rises sharply during 2020–2022, consistent with the pandemic disruption and sovereign default, and recedes during the post-2023 IMF-supervised recovery.

Table 8: Forecast Accuracy Across Horizons: RMSE (Balance of Payments Specification)

Model	$h = 1$	$h = 3$	$h = 6$	$h = 12$
MS-VAR	**315.2**	**679.5**	1150.7	2050.4
DMA	321.7	694.2	1143.8	2031.5
MS-VECM	323.4	842.8	1697.0	3249.8
DMS	326.0	701.8	**1049.8**	2018.3
LSTM	479.7	713.9	1057.3	**1742.2**
XGBoost	886.1	1376.1	1869.4	2679.0
BoPIdentity	1136.4	1006.3	1169.6	2734.7
ARIMA	1249.9	1463.9	1834.0	2448.6
Naïve	1350.8	1599.9	1914.3	2541.6
BVAR	1361.5	1597.3	1888.6	2487.1
LocalLevelSV	1410.0	1661.9	1964.2	2569.3
VECM	1556.3	1601.3	1921.3	2650.5

Notes: BoP set: reserves, exports, imports, remittances, tourism ($k = 5$). DMA and DMS combine forecasts from all non-combination models in this specification. Test period: 2023–2025. Bold indicates best at each horizon.

(1,742.2), suggesting that both dynamic model selection and sequence models can capture longer-range dependencies that static regime-switching specifications miss.

The pattern is revealing. DMA and DMS succeed not by inventing new information, but by dynamically reweighting the constituent models. During periods when the MS-VAR dominates, DMA assigns it the lion's share of weight. When no single model clearly leads, the combination hedge pays off. The net result is a forecast that is nearly as accurate as the best individual model at short horizons and more robust across horizons.

The success of regime-aware methods on disaggregated BoP data points to a broader pattern. Accurate reserve forecasting benefits from separately modelling the distinct dynamics of each balance-of-payments component (exports, imports, remittances, tourism) and allowing those relationships to shift across economic states. The parsimonious specification, which aggregates exports and imports into a single trade balance, discards information about asymmetric crisis dynamics that regime-switching models can exploit.

5.6 Variable Set Robustness

Table 9: Best Model by Variable Set: RMSE at $h = 1$ (Test Period)

Variable Set	Best Model	RMSE	vs. Naïve (%)
Parsimonious	MS-VAR	311.8	−73.6
BoP	MS-VAR	315.2	−76.7
Monetary	XGBoost	307.1	−76.3
PCA	MS-VECM	444.5	−65.7
Full	XGBoost	387.9	−70.1

Notes: The "vs. Naïve" column reports percentage RMSE reduction relative to the naïve random walk in each specification. A strong interaction pattern is evident: MS-VAR dominates in parsimonious and BoP sets, while XGBoost dominates in monetary and full sets.

Table 9 reveals a notable interaction between model class and variable set. MS-VAR dominates in the parsimonious and BoP specifications (RMSE $\approx$ 312–315), while XGBoost dominates in the monetary and full specifications (RMSE $\approx$ 307–388). This is consistent with regime models preferring compact, structurally relevant systems where state-dependent dynamics are cleanly identified, while tree models can exploit broader nonlinear feature spaces in richer specifications. The PCA specification is best served by MS-VECM (444.5), suggesting that dimensionality reduction partially preserves the regime structure that switching models exploit.

At the same time, cross-variable-set comparisons inherit different effective estimation windows (Table 3). For example, BoP estimation starts later than the monetary specification because tourism and remittance coverage begins later. The horse race is therefore clean within each variable set, while cross-set contrasts are interpreted with this sample-alignment caveat.

6 Mechanism Analysis

6.1 Architecture versus Information: The 2×2 Factorial Decomposition

Table 10: Architecture-Information Factorial Decomposition: 2×2 RMSE Matrix

	Parsimonious	BoP	Row Mean
MS-VAR	311.8	315.2	313.5
XGBoost	640.6	886.1	763.4
Column Mean	476.2	600.7	

Notes: Architecture effect (MS-VAR advantage, averaged across varsets) ≈ 449.9 RMSE points. Information effect ≈ −124.5 (BoP is worse on average, driven by XGBoost). Interaction term, using $\hat{\delta} = (R_{\text{XGB,Parsim}} - R_{\text{XGB,BoP}}) - (R_{\text{MSVAR,Parsim}} - R_{\text{MSVAR,BoP}})$, is ≈ −242.1.

Table 10 presents the factorial decomposition. The architecture effect is dominant: MS-VAR outperforms XGBoost by approximately 450 RMSE points on average across variable sets. The information effect is negative on average (−124.5), indicating that the BoP specification does not uniformly improve forecasts. It helps regime-switching models marginally but degrades XGBoost performance, whose feature importance analysis (Section 6.5) reveals heavy reliance on autoregressive reserve features rather than BoP components. The interaction term ($\hat{\delta} \approx -242.1$) indicates that moving from Parsimonious to BoP degrades XGBoost much more than MS-VAR (245.5 RMSE points versus 3.4). This is consistent with BoP disaggregation being more exploitable by regime-switching architecture than by tree-based models.

6.2 Regime Characterisation

Table 11 reports regime diagnostics. Both regimes are highly persistent. The crisis regime has a self-transition probability of 0.943, implying an expected duration of 17.6 months. The accumulation regime is even stickier at 0.983, with an expected duration of 57.2 months. Classification certainty is near-perfect: the mean maximum state probability is 0.997, and 99.5% of observations are classified with probability above 0.90. Entropy is correspondingly low (0.012). Classification shares (30.1% / 69.9%) differ from ergodic shares (23.5% / 76.5%), consistent with finite-sample state composition differing from long-run transition-implied frequencies.

This sharp separation is economically intuitive. Reserve accumulation is gradual, driven by persistent BoP surpluses. Crisis depletion is rapid, driven by sudden stops and peg defence. The data distinguish cleanly between these states.

Table 11: MS-VAR Regime Characterisation: BoP Specification

	Regime 0 (Crisis)	Regime 1 (Accumulation)
Self-transition probability	0.943	0.983
Expected duration (months)	17.6	57.2
Share of observations	30.1%	69.9%
Classification Quality		
Mean max probability	0.997	
Share with Pr $\geq$ 0.90	99.5%	
Mean entropy	0.012	
Estimation Diagnostics		
Converged	No (max iterations)	
Log-likelihood	-38.28	
Observations (differenced)	193	

Notes: Transition probabilities from EM estimation on the BoP variable set. Expected duration $= 1/(1 - p_{ii})$ where p_{ii} is the self-transition probability; durations are computed from full-precision probabilities while table probabilities are rounded to three decimals. "Share of observations" reports smoothed-state classification frequencies, not ergodic long-run shares. The ergodic distribution implied by the transition matrix is approximately 23.5% (Regime 0) and 76.5% (Regime 1).

Table 12: MS-VAR Multi-Start Stability Check (BoP Specification)

Metric	Estimation Stability (20 starts)	Forecast Stability ($h = 1$ test)
Convergence	20 / 20 runs converged	20 / 20 runs evaluated
Log-likelihood	mean -48.6; median -38.3	–
p_{00} / p_{11}	mean 0.920 / 0.934	–
Expected duration (months)	median 17.6 / 57.2	–
Classification certainty	mean max probability 0.987	–
RMSE across starts	–	mean 320.6 (SD 9.6)
RMSE range across starts	–	309.5 to 345.1
Improvement vs. Naïve (RMSE)	–	74.5% to 77.1%

Notes: Each start perturbs 5% of the baseline volatility-threshold initial state assignments and re-estimates the BoP MS-VAR with a 200-iteration EM cap. Forecast stability is computed from rolling-origin $h = 1$ RMSE on the BoP effective test window (2023:01–2025:03; $n = 27$).

A caveat remains: unconstrained random initialisations can still reach lower-likelihood local optima. However, the seeded perturbation multi-start check in Table 12 shows that the economically motivated baseline solution is locally stable in both regime diagnostics and out-of-sample loss. We therefore report regime estimates together with explicit initialization and convergence diagnostics, rather than relying on a single EM run.

6.3 Impulse Response Analysis

Regime-conditional generalised impulse response functions reveal asymmetry in shock propagation. The reserve response to its own shock peaks at roughly 559 in the accumulation regime, three times the crisis-regime peak of approximately 173. Across all target variables, the mean absolute peak delta is 97.3. Half-life differences average 9.0 months.

This asymmetry has an economic interpretation. During accumulation, reserve shocks propagate through policy responses (exchange rate management, import liberalisation) that amplify the initial impact. During crisis, binding constraints such as depleted reserves, capital controls, and suspended convertibility truncate shock propagation. Linear models average across these regimes. The MS-VAR does not, which contributes to its forecasting advantage.

6.4 Information Loss Under Aggregation

Table 13: Information Loss from Flow Aggregation

Period	Cancellation Index	Information Loss	RMSE Improvement	Share Helped
All	0.063	93.8%	−1.68%	57.1%
Crisis (2020–2022)	0.074	92.6%	−1.68%	57.1%
Tranquil (2023–2025)	0.059	94.1%	−2.92%	57.1%

Notes: Cancellation Index = $|A_t|/\sum|x_{jt}|$ where A_t is the aggregate and x_{jt} are gross flows. Information Loss = 1 − CI. RMSE Improvement = mean percentage change when moving from aggregated to disaggregated specification. Share Helped = proportion of models for which disaggregation reduces RMSE.

Table 13 reports the empirical information loss analysis. The mean cancellation index is 0.063, implying that approximately 94% of the information in gross BoP flows is destroyed by aggregation into net flows. This cancellation arises because exports and imports are both large and move in partially offsetting directions. During the crisis, export growth of 14.3% coexisted with import contraction of 26%, producing modest net changes that mask the underlying gross flow dynamics.

The RMSE improvement from disaggregation is modest in percentage terms (−1.68% to −2.92%) and model-dependent. Disaggregation helps regime-switching models (BVAR, MS-VECM) but hurts others. XGBoost shows a −20% RMSE degradation when moving

to the BoP specification. This asymmetric pattern is consistent with the architecture-information interaction identified in the 2×2 factorial decomposition.

6.5 Feature Importance

XGBoost feature importance analysis reveals that the model relies overwhelmingly on autoregressive reserve features rather than BoP components. In the BoP specification, the 3-month moving average of reserves alone accounts for approximately 75% of total feature importance. The next four features (reserve lags at horizons 1–3 and rolling standard deviation) contribute a further 13%. Disaggregated BoP variables (remittances, exports, imports, tourism) collectively account for less than 5%.

This finding explains why XGBoost gains little from moving to the BoP specification: it effectively treats the BoP variables as noise. The MS-VAR framework, by contrast, is structurally designed to model the joint dynamics of all system variables. Its regime-switching parameters capture state-dependent correlations between BoP components and reserves that the tree model cannot efficiently exploit.

6.6 Crisis Segment Performance and the Meese-Rogoff Puzzle

Table 14: Crisis Segment Performance: Best Models by Period

Segment	Best Model	RMSE	Naïve RMSE
Crisis (2020–2022, BoP validation)	MS-VAR (BoP)	518.2	1345.5
Post-default (2023–2025, BoP test)	MS-VAR (BoP)	315.2	1350.8
All pooled (2020–2025, BoP)	MS-VAR (BoP)	438.5	1338.0

Notes: RMSE values are BoP-specification $h = 1$ results from unified rolling-origin outputs. "All pooled" combines validation and test forecasts; effective pooled window is 2020:02–2025:03 due BoP data availability.

The results in Table 14 indicate that the Meese-Rogoff puzzle does not hold in its strong form for reserves. MS-VAR outperforms the naïve random walk during the 2020–2022 validation crisis segment: crisis RMSE of approximately 518 versus naïve crisis RMSE of 1,345, a 61.5% improvement. This suggests that regime-switching models, when applied to appropriately disaggregated data, can capture reserve depletion dynamics during sudden stops. The result extends beyond the standard Meese-Rogoff finding, which was formulated for exchange rate models applied to aggregated fundamentals.

However, the puzzle retains a weaker form. Among non-regime-switching models, the naïve benchmark remains competitive during crisis periods. None of ARIMA, VECM, BVAR, or XGBoost reliably beats it in the BoP crisis-validation segment. Classical and machine-learning models without explicit regime structure remain vulnerable to abrupt distributional breaks.

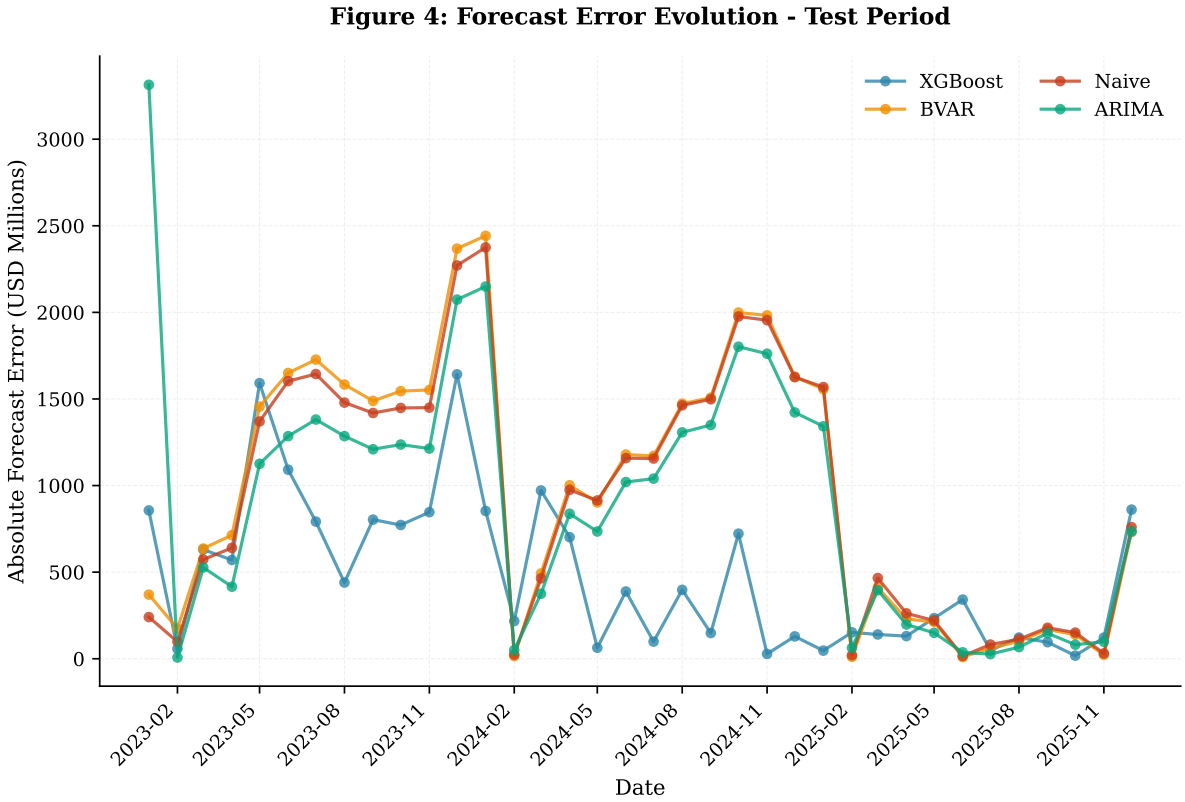

Figure 3: Absolute forecast errors over the test period (2023–2025) for selected models. MS-VAR maintains consistently lower errors, though all models exhibit episodic spikes corresponding to months of large reserve movements.

7 Density Forecast Evaluation

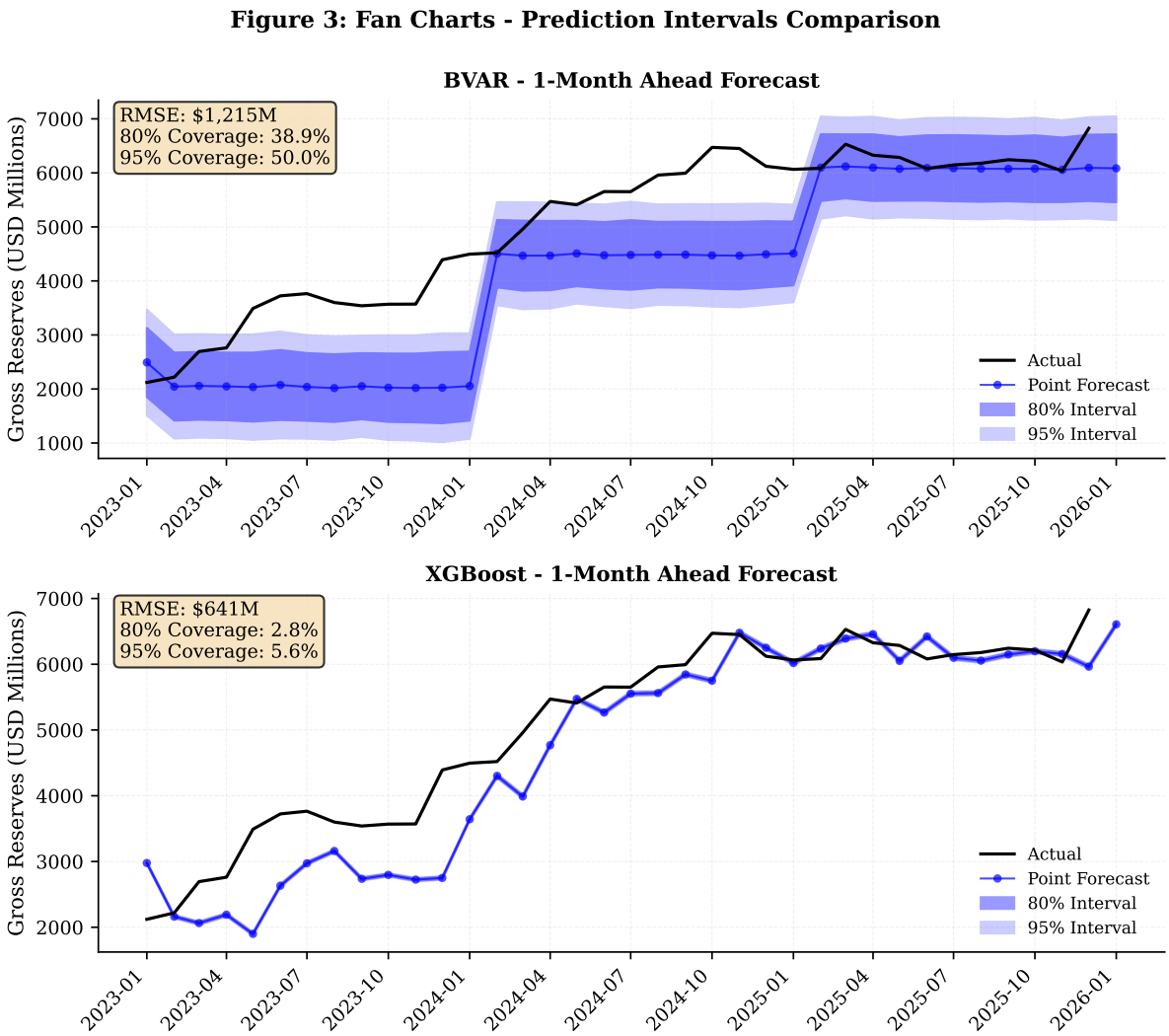

Figure 4: Prediction interval fan charts for BVAR (left) and XGB-Quantile (right). XGB-Quantile produces sharper, better-calibrated intervals; BVAR intervals are wider but exhibit substantial under-coverage relative to nominal levels.

Table 15 evaluates probabilistic forecasting performance. XGB-Quantile is the clear winner, with CRPS of 241.0 and the best coverage rates: 72.2% at 80% nominal and 97.2% at 95% nominal. Classical specifications (ARIMA, VECM) and BVAR exhibit poor coverage (36–50% at 80% nominal), suggesting that their estimated densities substantially underestimate forecast uncertainty—likely because parameter estimates are unstable across regimes. The XGBoost point-forecast model produces extremely narrow implicit intervals (2.8%/5.6% coverage). Good point forecasts, in short, do not guarantee good density forecasts.

The implication for policy is a two-model approach: regime-switching models for point forecasting and scenario analysis, complemented by XGB-Quantile or conformal prediction methods for uncertainty quantification.

Table 15: Density Forecast Evaluation: Parsimonious Specification, $h = 1$

Model	CRPS	Log Score	Coverage_{80} (%)	Coverage_{95} (%)
XGB-Quantile	**241.0**	–	**72.2**	**97.2**
XGBoost	466.4	−928.95	2.8	5.6
ARIMA	734.2	−10.12	41.7	50.0
Naïve	761.5	−9.88	44.4	52.8
VECM	782.4	−10.30	36.1	52.8
BVAR	792.1	−10.18	38.9	50.0

Notes: Coverage_{80} and Coverage_{95} = empirical coverage rates of 80% and 95% prediction intervals. Bold indicates best. Models without probabilistic output are marked with dashes.

8 Scenario Analysis and Robustness

8.1 Scenario Analysis

Table 16: MS-VARX Scenario Analysis: Reserve Projections (Parsimonious)

Scenario	End Level	Δ (USD m)	Δ (%)
Combined Upside	6837.7	12.7	0.19
Baseline	6755.7	−69.3	−1.02
Oil Price Shock	6724.9	−100.1	−1.47
Export Shock (−15%)	6718.7	−106.3	−1.56
IMF Tranche Delay	6624.5	−200.5	−2.94
LKR Depreciation 10%	6591.7	−233.3	−3.42
Combined Adverse	6509.7	−315.3	−4.62
LKR Depreciation 20%	6427.8	−397.2	−5.82

Notes: Projections based on MS-VARX model with parsimonious specification. 12-month horizon (2025:01–2025:12). Baseline from 2024:12 level of 6,825.0. Remittance and tourism shocks have zero impact under this specification as those variables are excluded; their effects propagate in BoP and fuller specifications.

Table 16 presents component-level scenario projections over a 12-month horizon. The baseline projects reserves at 6,755.7 USD million by end-2025, a modest 1.02% decline. Under a combined adverse scenario (LKR depreciation, export shock, oil price spike, IMF disbursement delay), reserves decline to 6,509.7 million (−4.62%), while a 20% currency depreciation alone produces a 5.82% decline. IMF disbursement delays prove particularly consequential (−2.94%), underscoring the sensitivity of the recovery trajectory to programme timing.

Variable set sensitivity analysis reveals that the full specification is most vulnerable to adverse shocks (9.97% decline under combined adverse in the full specification), whilst the parsimonious specification is most stable (4.62%). The BoP specification uniquely captures remittance decline impacts (−3.78%) and tourism recovery effects (+0.89%)

that are invisible in the parsimonious system.

8.2 Robustness to Train/Test Split

Table 17: Split Robustness: RMSE and Rank by Train/Validation Partition (BoP, $h = 1$)

	RMSE by Split			Rank Statistics	
Model	Baseline	Early	Late	Avg. Rank	Rank SD
MS-VAR	315.2	275.9	260.6	1.00	0.00
DMA	321.7	282.9	264.5	2.33	0.58
MS-VECM	323.4	291.4	268.9	3.33	0.58
DMS	326.0	285.4	260.6	2.67	1.53
XGBoost	886.1	1115.0	623.5	5.00	0.00

Notes: Baseline split: train to 2019-12, validate to 2022-12. Early split: train to 2018-12, validate to 2021-12. Late split: train to 2020-12, validate to 2023-12. DMS ties MS-VAR in the late split (260.6). The split-robust evaluation set excludes LSTM and BoPIdentity (configuration flags 'include_lstm = false', 'include_bop = false'); the table therefore reports the top five models within that evaluated set. Remaining evaluated models (ARIMA, Naïve, BVAR, VECM) have RMSE > 1,200 in all splits.

Table 17 addresses whether the model ranking is an artefact of a particular train/validation/test partition. Under three split definitions (baseline: train to 2019-12, validate to 2022-12; early: 2018-12/2021-12; late: 2020-12/2023-12), MS-VAR holds rank 1 in every case. The corresponding RMSEs are 315.2, 275.9, and 260.6. Its average rank is 1.00 with zero rank variance.

DMA is the most stable runner-up (average rank 2.33, rank SD 0.58). DMS ties MS-VAR in the late split at 260.6 but is less stable overall (average rank 2.67, rank SD 1.53). The gap between the top-four cluster and XGBoost widens in the early split, where XGBoost deteriorates to 1,115.0, nearly four times the MS-VAR error.

These results support two claims. First, the gains from regime-switching architecture are genuine, not driven by a favourable partition. Second, stability and accuracy are distinct properties. Some simpler models (e.g., Naïve) are relatively stable across splits but at worse loss levels. The regime-aware models are both accurate and stable, which is uncommon in macroeconomic forecasting comparisons [Rossi, 2013].

Split choice does, however, materially affect regime diagnostics. The training-validation subsample converged where the full-sample estimation often did not, and estimated regime durations differ across splits. Production forecasting systems should report results across multiple configurations and flag cases where regime diagnostics are qualitatively different.

9 Conclusion

This paper compares classical econometric, Bayesian, regime-switching, and machine learning models for forecasting Sri Lanka's foreign exchange reserves. The evaluation spans five variable sets, multiple forecast horizons, and a dataset encompassing stable accumulation, pandemic disruption, sovereign default, and IMF-supervised recovery. Beyond the model comparison, the analysis introduces a factorial architecture-versus-information decomposition and provides analytical evidence for the sources of regime-switching forecast advantage.

The central result is the strong performance of regime-aware methods. MS-VAR achieves RMSE reductions of roughly three-quarters relative to the naïve benchmark. DMA and DMS perform nearly as well, and the 90% Model Confidence Set retains all four regime-aware specifications while eliminating all others. This ranking is stable across three alternative train/test splits. The 2×2 factorial decomposition shows that the advantage is primarily architectural: regime-switching structure accounts for the bulk of the improvement over gradient-boosted trees. The interaction term indicates that architecture and information content are complements for regime-switching models but substitutes for tree-based methods.

The information-loss analysis demonstrates that aggregating balance-of-payments flows destroys the majority of the available signal through cancellation of offsetting gross flows. Regime characterisation reveals highly persistent states with near-perfect classification certainty. Impulse response analysis shows regime asymmetry in shock propagation consistent with distinct crisis and recovery dynamics.

The Meese-Rogoff puzzle does not hold in its strong form for reserves. MS-VAR outperforms the naïve benchmark even during the acute crisis segment. A weaker form persists: among non-regime-switching models, the random walk remains competitive during crisis periods. XGBoost dominates in the monetary and full specifications, suggesting that tree models can exploit broader nonlinear feature spaces when dimensionality is sufficient. However, their reliance on autoregressive reserve features limits their ability to exploit structural BoP information.

9.1 Limitations and Future Work

Several limitations circumscribe these findings. First, although seeded multi-start checks indicate local stability around the baseline MS-VAR solution, broader random initialisations can still locate inferior local optima. Regime estimation in short samples therefore remains non-trivial and should always be accompanied by reported convergence diagnostics. Second, LSTM robustness remains incompletely explored. Recurrent architectures are sensitive to sequence length, learning rate, and regularisation. More exhaustive tuning could alter the relative ranking of neural network approaches. Third, the single-country

design limits generalisability. Application to a cross-country panel of emerging markets, particularly those with different external vulnerability profiles, would test whether the MS-VAR advantage is specific to Sri Lanka or reflects a more general property. Fourth, real-time data vintage effects are not explicitly modelled. The use of revised data may overstate forecast accuracy relative to what would have been achievable in real time.

Several avenues for future research emerge. Conformal prediction methods offer a principled approach to combining regime-switching point-forecast accuracy with distribution-free uncertainty quantification. Such methods could resolve the gap between MS-VAR's point forecast advantage and XGB-Quantile's superior density calibration. Ensemble methods combining regime-switching models with machine learning via stacking could exploit complementary strengths: structural regime identification from MS-VAR and flexible nonlinear feature extraction from tree-based methods. Finally, extending the information-loss framework to other macroeconomic aggregates (GDP components, monetary aggregates) would test whether the cancellation documented here for balance-of-payments flows is a broader phenomenon.

Despite these limitations, the evidence consistently favours regime-aware approaches. Explicitly modelling regime dynamics in disaggregated balance-of-payments systems appears strongly beneficial for reserve forecasting in crisis-prone emerging markets, though the single-country design means this conclusion should be tested in other settings. Central banks confronted with reserve surveillance should consider pairing regime-switching models for scenario analysis with complementary probabilistic methods for uncertainty quantification.

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A Estimation Details and Metric Definitions

A.1 Classical Models

ARIMA models with exogenous regressors are estimated via maximum likelihood, with lag order selected via AIC across ARIMA(p, d, q) specifications with $p, q \in \{0, 1, 2, 3\}$ and $d \in \{0, 1\}$. VAR and VECM models are estimated via full-information maximum likelihood with lag order selected by AIC (maximum 4 lags). Cointegration rank is determined by the Johansen trace and maximum eigenvalue tests.

A.2 Bayesian VAR

The BVAR Minnesota prior is centred on a random walk with hyperparameters: overall shrinkage $\lambda \in [0.1, 1.5]$, lag decay $\mu \in [0.1, 0.8]$, and own-lag variance $\phi \in [0.1, 0.95]$, selected via grid search on in-sample BIC. The posterior is simulated via Gibbs sampling with 10,000 draws (2,000 burn-in).

A.3 Machine Learning

XGBoost: 200 boosting rounds (300 for BoP), learning rate $\eta = 0.1$, tree depth 3–5, minimum child weight 1, subsample ratio 0.8–1.0, column subsample 0.8, time-series cross-validation. Features include lagged dependent variable (lags 1–12), lagged exogenous variables (lags 1–6), and rolling 3- and 6-month means and volatilities.

LSTM: single recurrent layer of 64 units, input sequence length 12, Adam optimiser (learning rate 0.001), dropout 0.2, batch size 32, maximum 100 epochs with early stopping (patience 10).

A.4 Accuracy Metrics

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{1}$$

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \tag{2}$$

$$\text{MAPE} = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{3}$$

$$\text{MASE} = \text{MAE}\Big/\frac{1}{n-1}\sum_{t=2}^{n}|y_t - y_{t-1}| \tag{4}$$

The asymmetric policy loss function is defined as:

$$L(e_t) = \begin{cases} 2 \cdot e_t^2 & \text{if } e_t < 0 \text{ (under-prediction)} \\ e_t^2 & \text{if } e_t \geq 0 \text{ (over-prediction)} \end{cases}$$

Probabilistic forecast metrics:

$$\text{CRPS} = \frac{1}{n}\sum_{t=1}^{n}\int_{-\infty}^{\infty}[F_t(z) - \mathbf{1}_{z \geq y_t}]^2\, dz \tag{5}$$

$$\text{Log score} = \log p_t(y_t) \tag{6}$$

The Diebold-Mariano statistic is $\text{DM} = \bar{d}/\sqrt{2\pi f_d(0)/n}$, which is asymptotically standard normal under the null of equal expected loss.